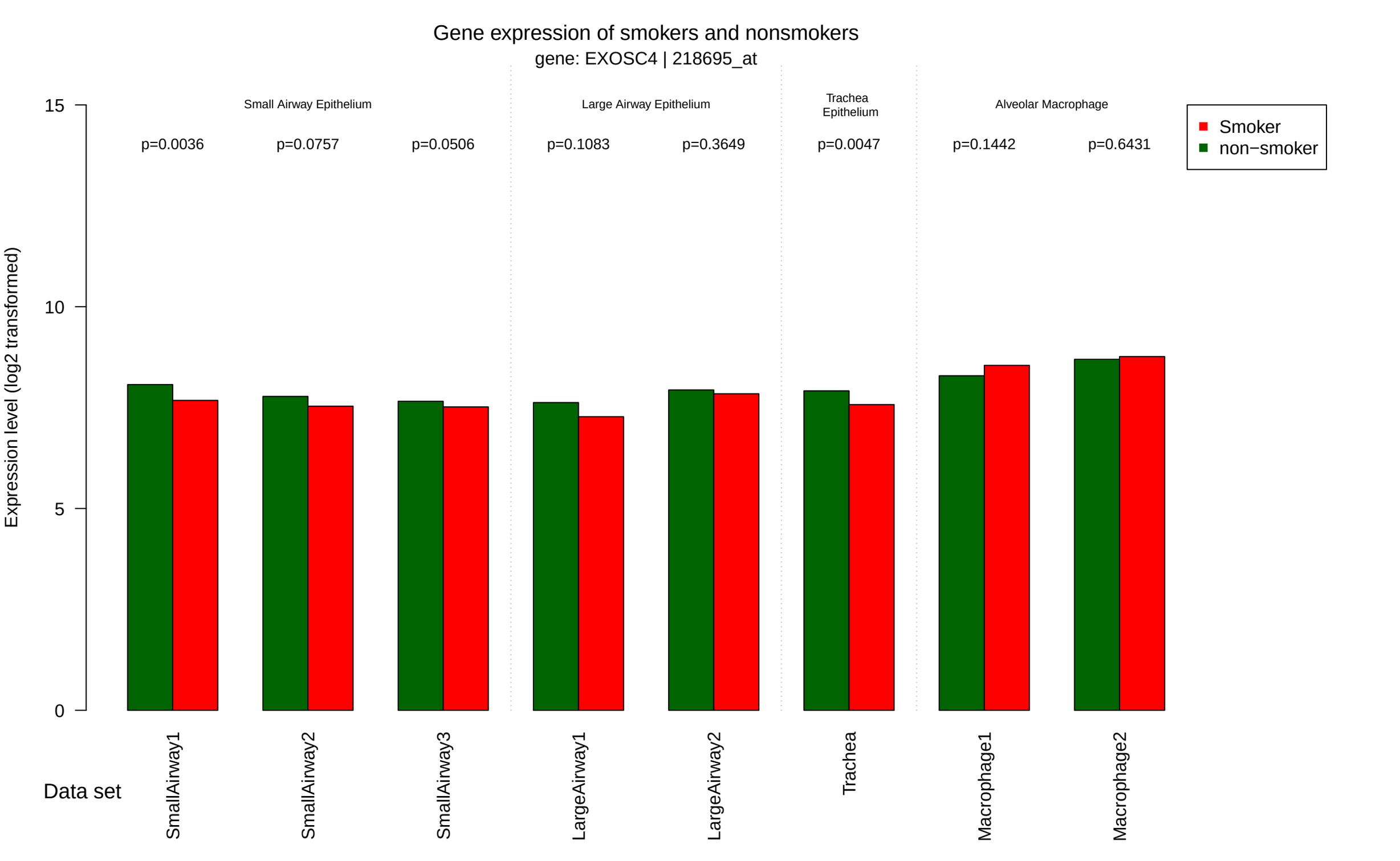
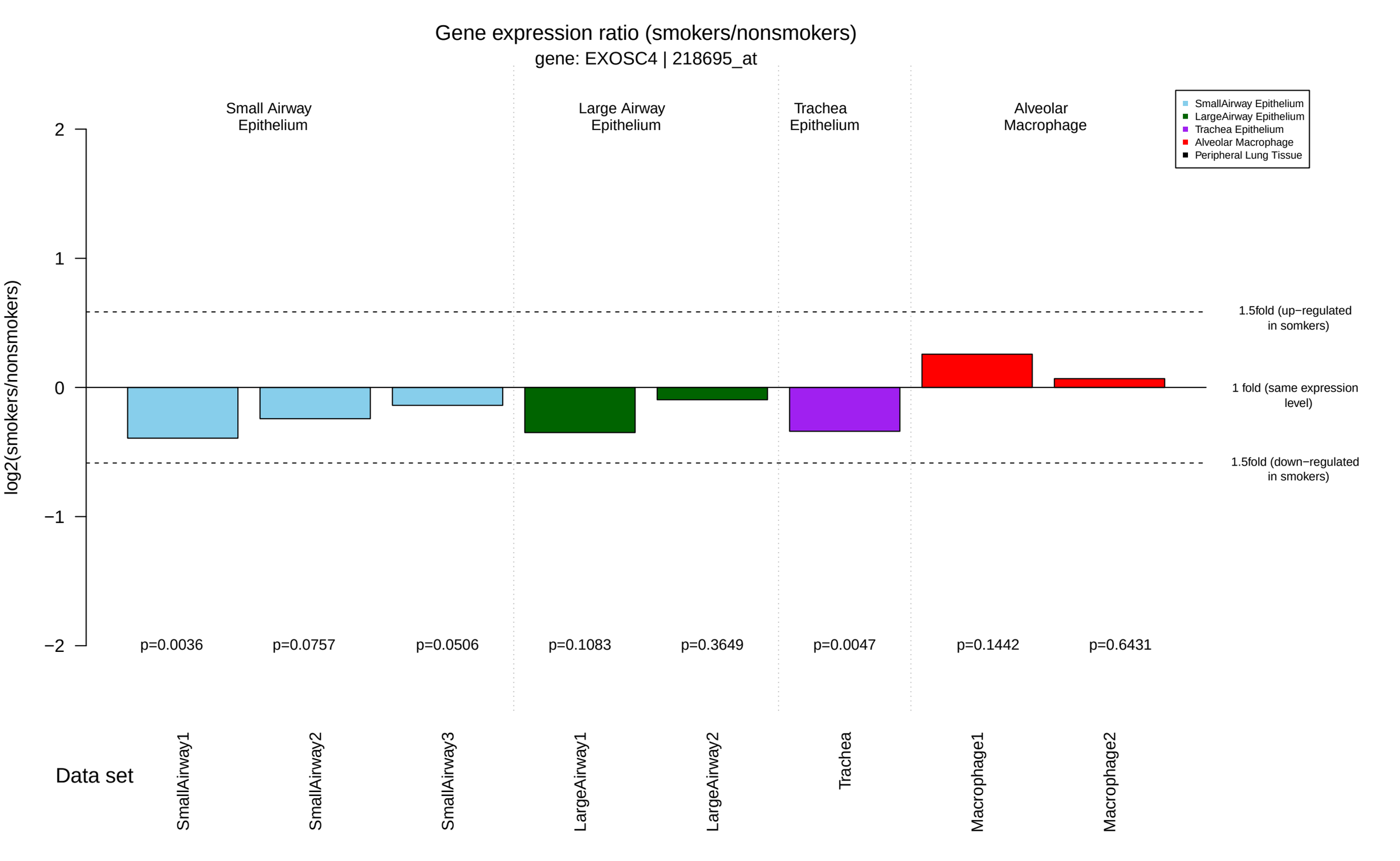
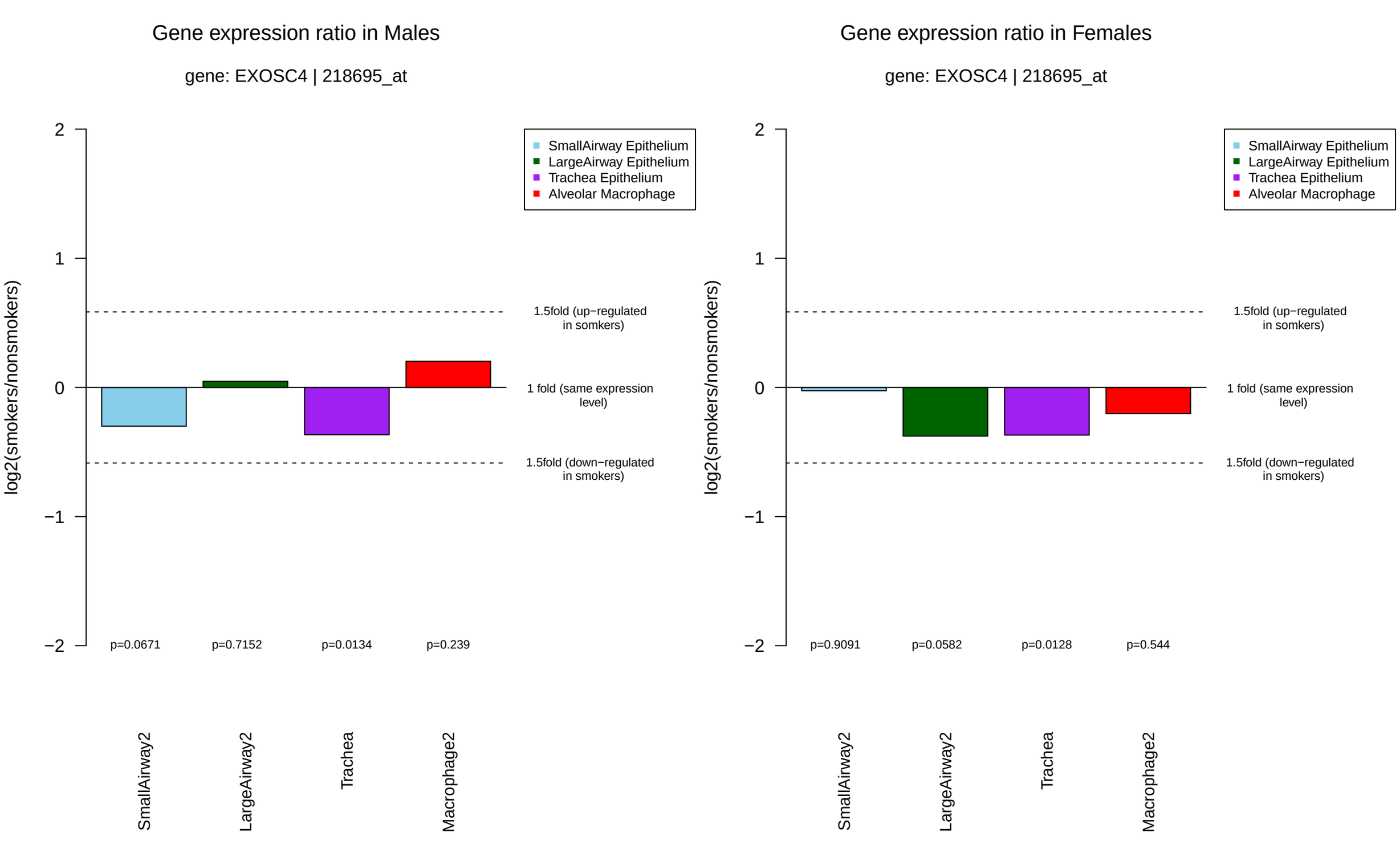
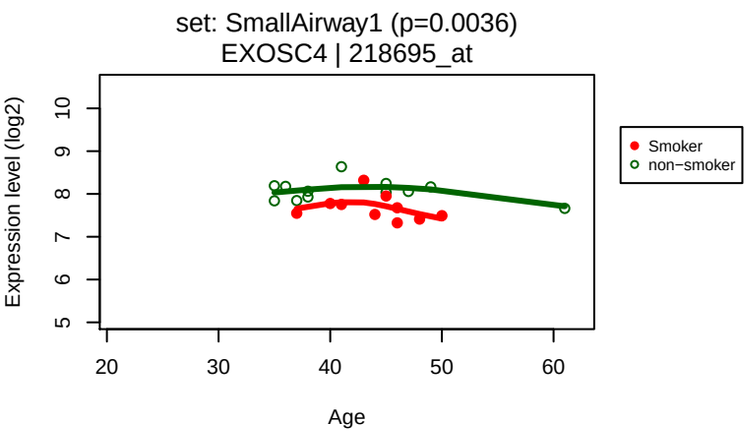
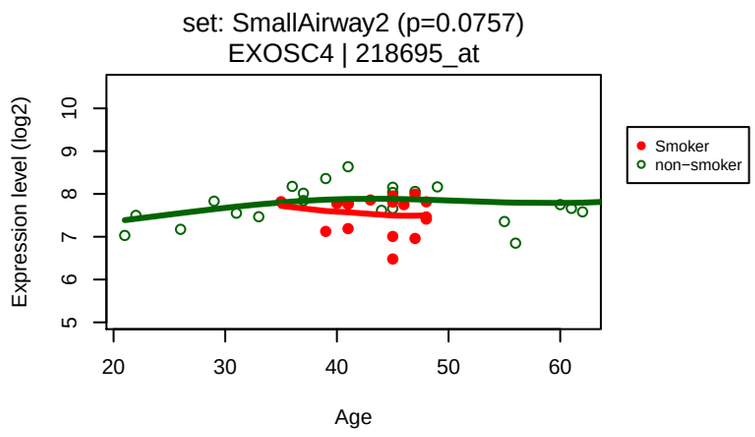
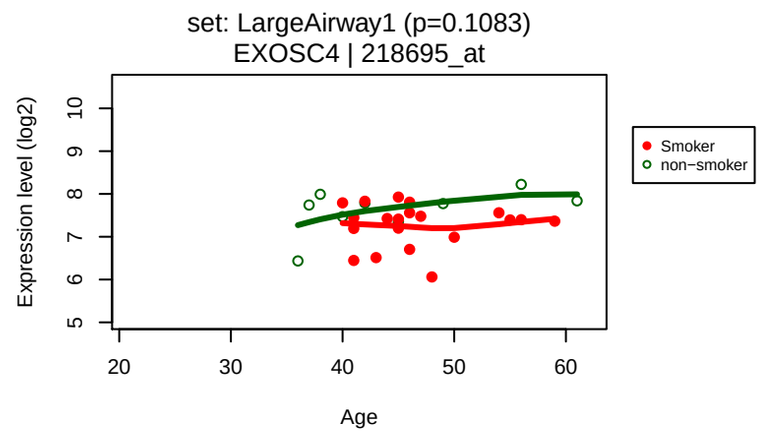

Gene expression of smokers and nonsmokers
gene: EXOSC4 | 218695_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0036
p=0.0757
p=0.0506
p=0.1083
p=0.3649
p=0.0047
p=0.1442
p=0.6431
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

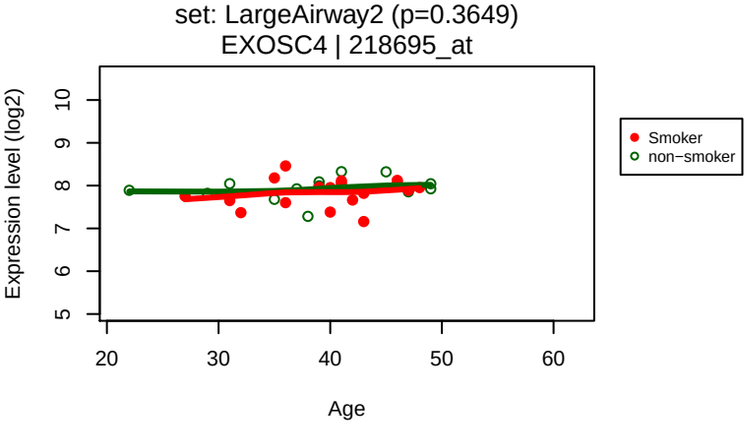

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC4 | 218695_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0036
p=0.0757
p=0.0506
p=0.1083
p=0.3649
p=0.0047
p=0.1442
p=0.6431
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

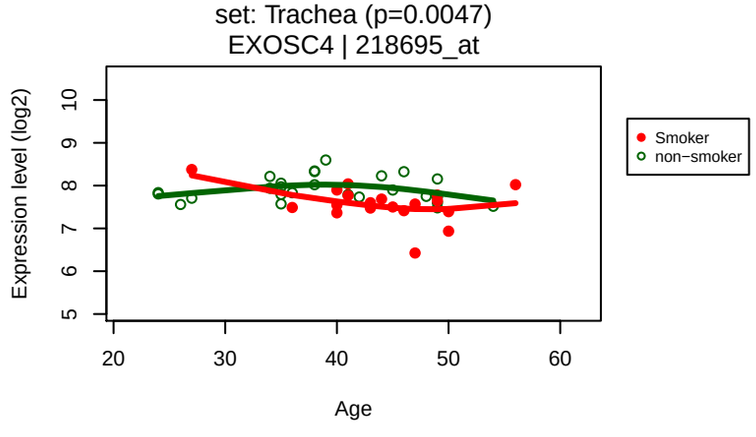

Gene expression ratio in Males
gene: EXOSC4 | 218695_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0671
p=0.7152
p=0.0134
p=0.239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC4 | 218695_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9091
p=0.0582
p=0.0128
p=0.544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

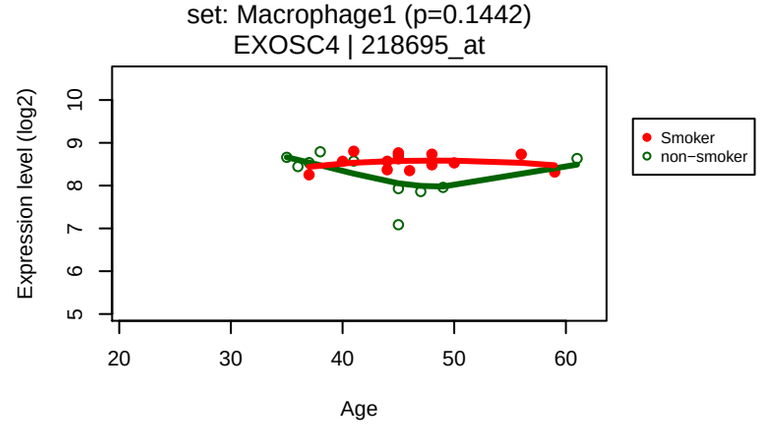

set: SmallAirway1 (p=0.0036)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker

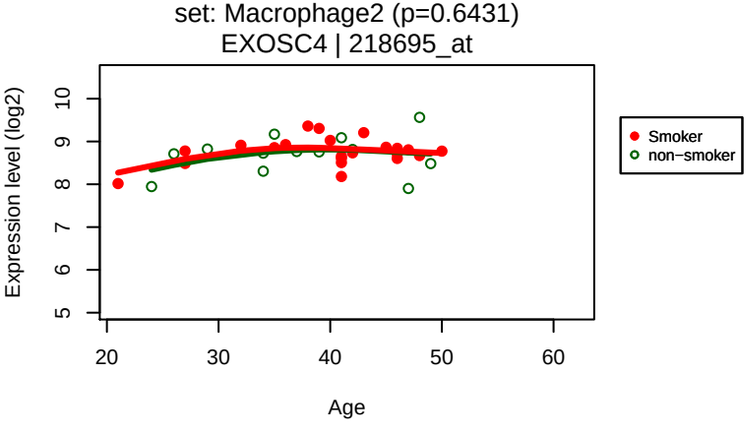

set: SmallAirway2 (p=0.0757)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker

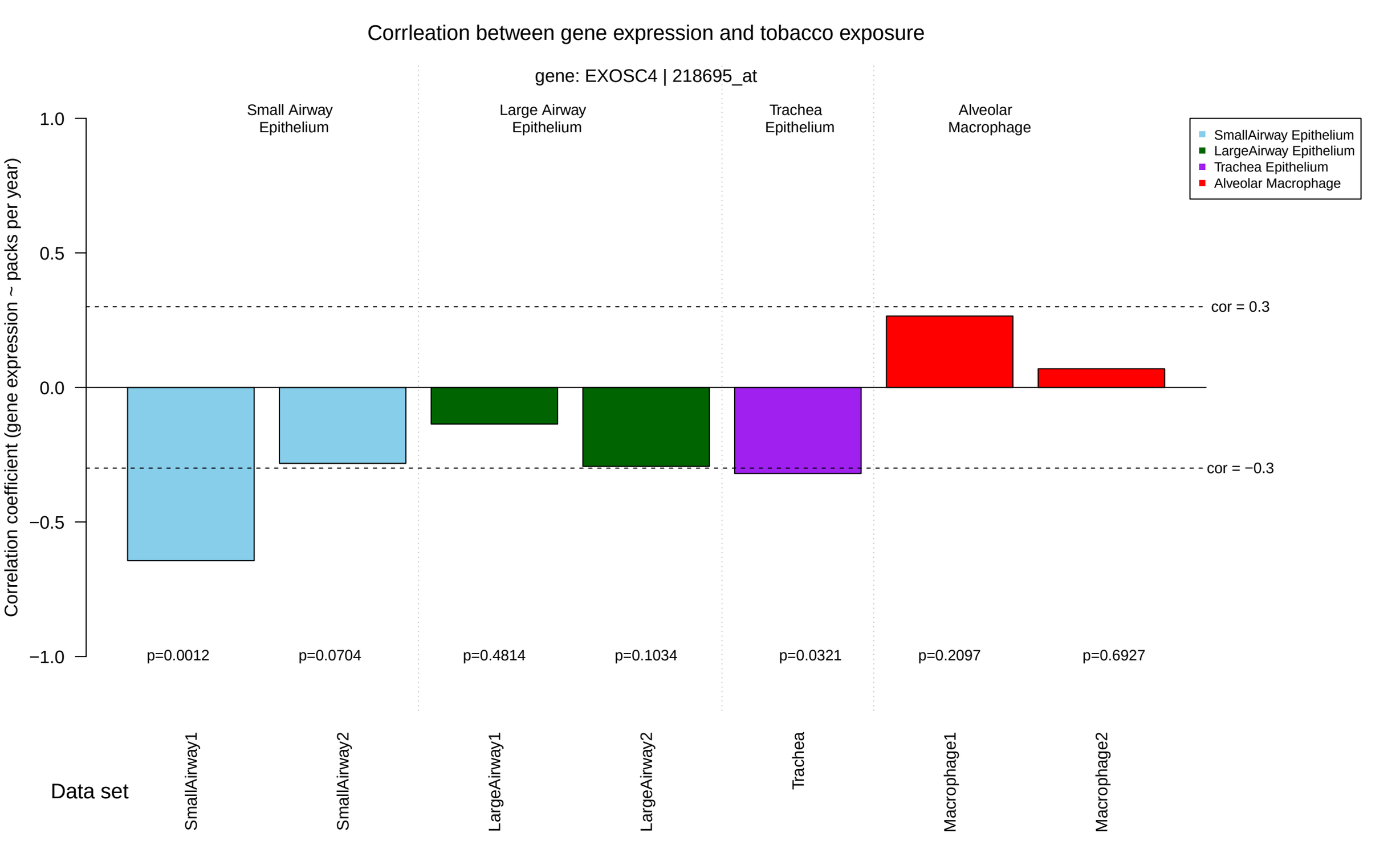

set: LargeAirway1 (p=0.1083)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3649)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0047)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1442)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6431)
EXOSC4 | 218695_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC4 | 218695_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0012
p=0.0704
p=0.4814
p=0.1034
p=0.0321
p=0.2097
p=0.6927
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2